

BSDS (2025 Semester II): Statistics II

Endsem: Part A

NAME :	ROLL :
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Time: 90 minutes

Total marks: 52

Maximum attainable marks: 45

All the rough works should be done in the space provided in the question paper.

Q.I Each of the following questions has one or more correct options. Identify all of them. $6 \times 4 = 24$

1. Let $X \sim F_\theta$ where $F_\theta \equiv \text{Uniform}(0, \theta)$. Indicate the correct statements below.

A. F_θ belongs to a scale family. ☒

B. F_θ belongs to a location family. ☒

C. F_θ belongs to an exponential family. ☒

D. The distribution of $1/X$ belongs to a scale family. ☒

1. A. $X \sim \text{Unifam}(0, \theta)$. Define $W = \frac{X}{\theta}$. Then $W \in (0, 1)$ and $J_{XW} = \frac{\partial x}{\partial w} = \theta$.

A.1 So, $f_W(w) = \frac{1}{\theta} \cdot \theta = 1$; $0 < w < 1$. Thus, $W \sim \text{Uniform}(0, 1)$, which is free of θ . Therefore, $X \sim \text{scale family}$.

1. B. Define $Y = X - c$; $c \in \mathbb{R}$. Then $Y \sim \text{Uniform}(-\theta - c, \theta - c)$, which is not free of θ for any choice of c . So, X does NOT belong to location family.

1. C. Since the support of X is $(0, \theta)$, which depends on parameter θ , X does NOT belong to exponential family.

1. D. Let $Z = 1/X$, then $\theta Z = \theta/X = 1/W$ has a distribution free of θ (from A.)
So, Z belongs to scale family.

2. Consider that $X \sim \text{Exponential}(\lambda)$ distribution with expectation $1/\lambda$, and a constant $\alpha \in (0, 1/2)$. Let $Y = \exp\{-X\}$ and γ_α is the upper- α point of X .

A. The upper- α point of Y is $\exp\{-\gamma_\alpha\}$. ☒

B. The upper- α point of X is $\exp\{-\lambda\alpha\}$. ☒

C. The upper- α point of X is $-\{\log(1 - \alpha)\}/\lambda$. ☒

D. $E(X)$ is the upper- α point of X , where $\alpha = \exp\{-1\}$. ☒

A.2 2. A. Given $P(X > \gamma_\alpha) = \alpha = P(\bar{e}^X < \bar{e}^{\gamma_\alpha})$. So $\exp\{-\gamma_\alpha\}$ is lower- α point of $Y = \bar{e}^X$.

2. B. $P(X > \gamma_\alpha) = \alpha = \int_{\gamma_\alpha}^{\infty} \lambda \bar{e}^{-\lambda x} dx = \left[-\bar{e}^{-u} \right]_{\lambda \gamma_\alpha}^{\infty} = \bar{e}^{-\lambda \gamma_\alpha} \Rightarrow \gamma_\alpha = -\frac{1}{\lambda} \log \alpha \neq \exp\{-\lambda \alpha\}$

2. C. From B. $\gamma_\alpha = -\frac{1}{\lambda} \log \alpha \neq -\frac{1}{\lambda} \log(1 - \alpha)$ as $\alpha < 1/2$.

2.D. $E(X) = \bar{\lambda}^{-1}$. $P(X > \bar{\lambda}) = \left[-e^{-u} \right]_1^{\infty} = e^{-1}$.

3.A. Let G_{1n} and g_{1n} be the CDF and PDF of $T_{1n} = X_{(n)}$.

A.3 Then $G_{1n}(t) = [P(X_1 < t)]^n = \left[\int_{-\theta}^t \frac{du}{2\theta} \right]^n = \left(\frac{t+\theta}{2\theta} \right)^n$ for $-\theta < t < \theta$.

3. Let $X_1, \dots, X_n \stackrel{i.i.d.}{\sim} \text{Uniform}(-\theta, \theta)$ distribution. Consider three estimators of θ , $T_{1,n} = X_{(n)}$, $T_{2,n} = -X_{(1)}$ and $T_{3,n} = Y_{(n)}$, where $Y_i = |X_i|$ for $i = 1, \dots, n$.

A. $E(T_{1,n}) \rightarrow \theta$ as $n \rightarrow \infty$. ✓

B. $E(T_{2,n})$ over-estimates θ . ✗

C. $E(T_{3,n}) \rightarrow \theta$ as $n \rightarrow \infty$. ✓

D. The squared bias of $T_{3,n}$ is always smaller than that of $T_{1,n}$ or $T_{2,n}$. ✓

[Hint: Try to express $T_{3,n}$ as a function of $T_{1,n}$ and $T_{2,n}$.]

Thus, $g_{1n}(t) = \frac{n(t+\theta)^{n-1}}{(2\theta)^n}$; $-\theta < t < \theta$, and θ

$$E(T_{1n}) = \int_{-\theta}^{\theta} t \frac{n(t+\theta)^{n-1}}{(2\theta)^n} dt = \frac{n}{(2\theta)^n} \left[\int_{-\theta}^{\theta} (t+\theta)^n dt - \theta \int_{-\theta}^{\theta} (t+\theta)^{n-1} dt \right]$$

$$= \frac{n}{(2\theta)^n} \left[\frac{(2\theta)^{n+1}}{n+1} - \theta \frac{(2\theta)^n}{n} \right] = n \left[\frac{2\theta}{n+1} - \frac{\theta}{n} \right] = \theta \left(\frac{n-1}{n+1} \right) \rightarrow \theta \text{ as } n \rightarrow \infty.$$

3.B. $X_{(1)} > -\theta$ with prob. 1 $\Leftrightarrow T_{2n} = -X_{(1)} < \theta$ w.p. 1. So, T_{2n} UNDER-estimates θ .

3.C. Observe that, $T_{3n} = \max\{X_{(n)}, -X_{(1)}\} = \max\{T_{1n}, T_{2n}\}$.

Further, if $Z_i = -X_i$, $i=1, \dots, n$. Then $Z_i \stackrel{i.i.d.}{\sim} \text{Uniform}(-\theta, \theta)$ and $Z_{(n)} = -X_{(1)} = T_{2n}$.
So, $E(T_{2n}) = E(Z_{(n)}) = \theta \left(\frac{n-1}{n+1} \right) \rightarrow \theta$.

4. Let $X_1, \dots, X_n \stackrel{i.i.d.}{\sim} \text{Bernoulli}(\theta)$ distribution. Consider the problem of testing $H_0 : \theta = 0.5$ against $H_1 : \theta > 0.5$, and the test function of a level- α test

$$\phi(x) = \begin{cases} 1 & \text{if } \sum_{i=1}^n x_i > c_\alpha, \\ \gamma & \text{if } \sum_{i=1}^n x_i = c_\alpha, \\ 0 & \text{otherwise,} \end{cases}$$

where $\gamma \in (0, 1)$. Let $\mathbf{y} = \{y_1, \dots, y_n\}$ be a realization of $\{X_1, \dots, X_n\}$.

A. If H_0 is rejected given $\mathbf{y}$ at 5% level of significance, then it must be rejected at 1% level of significance. ✗

B. Let $\sum_{i=1}^n y_i = c_\alpha$, and $p(\mathbf{y})$ be the p -value given the sample $\mathbf{y}$. Then H_0 will be accepted based on the $p(\mathbf{y})$. ✓

C. Let $\sum_{i=1}^n y_i = c_\alpha$, then the test cannot be conducted without evaluating the p -value. ✗

D. Let $\sum_{i=1}^n y_i = c_\alpha + 1$, then $p(\mathbf{y}) > \alpha$. ✗

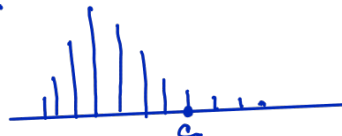
[3.C. cont...] Finally, $T_{3n} < \theta$ w.p. 1. So, $\theta \geq E(T_{3n}) \geq \max\{E(T_{1n}), E(T_{2n})\} \rightarrow \theta$
As $E(T_{1n}) \rightarrow \theta$ as $n \rightarrow \infty$, $i=1, 2$. $E(T_{3n}) \rightarrow \theta$ as $n \rightarrow \infty$ by Sandwich thm.

3.D. By (*), $\theta - E(T_{3n}) \leq \min\{\theta - E(T_{1n}), \theta - E(T_{2n})\}$. So, squared bias of T_{3n} is less.

A.4

4.A. At 5% level critical region is larger than that at 1% level.
So, a sample can be rejected at 5% level but
accepted at 1% level.

4.B. Since $\gamma \in (0,1)$, $P_{H_0}(\sum_{i=1}^n X_i > c_\alpha) < \alpha$ and $P_{H_0}(\sum_{i=1}^n X_i \geq c_\alpha) > \alpha$,
but the p-value of the sample $\underline{y}$ is $P_{H_0}(\sum_{i=1}^n X_i \geq \sum_{i=1}^n y_i) = P_{H_0}(\sum_{i=1}^n X_i \geq c_\alpha) > \alpha$.
So, the test must be accepted.



5. Let X be a random variable taking values in $\{-3, \dots, 3\}$. Consider the testing problem
 $H_0: X \sim f_0$ against $H_1: X \sim f_1$, where

x	-3	-2	-1	0	1	2	3
$f_0(x)$	0.20	0.15	0.10	0.05	0.10	0.15	0.25
$f_1(x)$	0.05	0.10	0.15	0.40	0.15	0.10	0.05

Consider the most powerful level- α test ϕ^* by Neyman-Pearson's (NP) lemma, and let
 $\alpha = 0.05$. Then indicate which of the following statements are true.

A. $\phi^*(X)$ is a randomized test. ☒

B. $\phi^*(X) = \mathbb{I}(X = 0)$, where $\mathbb{I}$ is the indicator function. ☒

C. The probability of type II error for $\phi^*(X)$ is 0.40. ☒

D. $\phi^*(X)$ is unbiased. ☒

4.D. For the sample $\underline{y}$ satisfying $\sum_{i=1}^n y_i = c_\alpha + 1$, p-value is
 $P(\underline{y}) = P(\sum_{i=1}^n X_i \geq \sum_{i=1}^n y_i) = P(\sum_{i=1}^n X_i \geq c_\alpha + 1) = P(\sum_{i=1}^n X_i > c_\alpha) < \alpha$.

A.5

5.A. By the NP lemma, the MP is of the form $\phi^*(x) = \begin{cases} 1 & \text{if } \lambda(x) > k \\ 0 & \text{if } \lambda(x) < k \end{cases}$; and satisfies
 $E_{H_0}(\phi^*(X)) = \alpha = 0.05$, where $\lambda(x) = \frac{f_1(x)}{f_0(x)}$.

Here,

x	-3	-2	-1	0	1	2	3
$\lambda(x)$	1/4	2/3	3/2	8	3/2	2/3	1/5

 Thus, for any choice of $k \in (3/2, 8)$,
the following things happen:

i. $\phi^*(x) = \begin{cases} 1 & \text{if } x = 0 \\ 0 & \text{otherwise} \end{cases}$; as $P_{H_0}(\lambda(X) = k) = 0$; ii. $E_{H_0}(\lambda(X) \geq k) = P_{H_0}(X = 0) = \frac{f_0(0)}{f_0(0)} = 0.05$

So, the MP test ϕ^* is NOT randomized.

5.B. TRUE (obvious from A.)

5.C. $P(\text{Type II error of } \phi^*) = P_{H_1}(X \neq 0) = 0.6 \neq 0.4$. (5.D) Power of $\phi^* = 1 - 0.6 = 0.4$
0.05 = size. So, ϕ^* is unbiased.

6. Let $X_n, n \in \mathbb{N}$, be a sequence of independent random variables such that $X_n \sim \text{Binomial}(m, 1/n)$,
and $Y_n = \sum_{j=1}^n X_j$. Then which of the following statements are correct?

A. Y_n is distributed as $\text{Binomial}(m, 1)$. ☒

B. $X_n \xrightarrow{P} 0$ as $n \rightarrow \infty$. ☒

C. $X_n \xrightarrow{L^2} 0$ as $n \rightarrow \infty$. ☒

D. $\text{var}(Y_n) \rightarrow 0$ as $n \rightarrow \infty$. ☒

A.6

6.A. FALSE. The support of $Y_n = \sum_{j=1}^n X_j$ is $\{0, \dots, nm\}$ but that of $\text{Bin}(m, 1)$ is $\{m\}$.

6.B. $E(X_n) = m/n \rightarrow 0$ as $n \rightarrow \infty$, m fixed; and $\text{var}(X_n) = \frac{m}{n}(1 - \frac{1}{n}) \rightarrow 0$ as $n \rightarrow \infty$.

Thus, $X_n \xrightarrow{P} 0$.

6.C. $E(X_n^2) = \text{var}(X_n) + E^2(X_n) = \frac{m}{n}(1 - \frac{1}{n}) + \frac{m^2}{n^2} \rightarrow 0$ as $n \rightarrow \infty$. So $X_n \xrightarrow{L^2} 0$.

6.D. $\text{var}(Y_n) = \text{var}(\sum_{j=1}^n X_j) = \sum_{j=1}^n \text{var}(X_j) + \sum_{j \neq j'} \text{cov}(X_j, X_{j'}) = \sum_{j=1}^n \text{var}(X_j) = n \frac{m}{n}(1 - \frac{1}{n})$

Q.II Write the answers to the following questions in one sentence.

$7 \times 4 = 28$

1. Let $X_1, \dots, X_n \stackrel{i.i.d.}{\sim} \text{Poisson}(\lambda)$ distribution. Consider the function of λ , $g(\lambda) = \exp\{-\lambda\}$. Find an unbiased estimator of $g(\lambda)$.

Observe that $g(\lambda) = \exp\{-\lambda\} = P(X_1 = 0)$.

Define $Y_1 = \mathbb{I}(X_1 = 0) = \begin{cases} 1 & \text{if } X_1 = 0 \\ 0 & \text{otherwise} \end{cases}$

Then $E(Y_1) = 1 \cdot P(X_1 = 0) = e^{-\lambda}$. Thus, Y_1 is an unbiased estimator of $g(\lambda)$.

[Hint: Express $g(\lambda)$ as $P(A_X)$ for some appropriate event A_X .]

2. Let $X_1, \dots, X_n \stackrel{i.i.d.}{\sim} \text{Pareto}(\xi_0, \alpha)$ distribution, with pdf

$$f_X(x) = \frac{\alpha \xi_0^\alpha}{x^{\alpha+1}}, \quad x > \xi_0 > 0, \quad \alpha > 0,$$

where ξ_0 is known. Find the MLE of α .

The log-likelihood of α is:

$$\ln(\alpha) = \log \left(\prod_{i=1}^n \frac{\alpha \xi_0^\alpha}{x_i^{\alpha+1}} \right) = n \log \alpha + n \alpha \log \xi_0 - (\alpha+1) \sum_{i=1}^n \log x_i$$

First order condition: $\frac{\partial \ln(\alpha)}{\partial \alpha} = 0 \Rightarrow \frac{n}{\alpha} + n \log \xi_0 - \sum_{i=1}^n \log x_i = 0$

(FOC)

$$\Rightarrow \hat{\alpha} = \left[\frac{\sum_{i=1}^n \log(x_i/\xi_0)}{n} \right]^{-1} \text{ is the solution of FOC.}$$

Next, $\frac{\partial^2 \ln(\alpha)}{\partial \alpha^2} = -\frac{n}{\alpha^2} < 0 \quad \forall \alpha$. Thus, $\hat{\alpha}$ is the MLE.

3. Let $X_n \sim N(\mu_n, \sigma_n^2)$ where $\mu_n \rightarrow 0$ and $\sigma_n^2 \rightarrow 0$, as $n \rightarrow \infty$. Show that $X_n \xrightarrow{L^2} 0$ as $n \rightarrow \infty$.

$$E(X_n^2) = \text{Var}(X_n) + E^2(X_n) = \sigma_n^2 + \mu_n^2 \rightarrow 0 \text{ as } n \rightarrow \infty \text{ by the given conditions.}$$

$$\text{Thus, } X_n \xrightarrow{L^2} 0.$$

4. Let $X_1, \dots, X_n \stackrel{i.i.d.}{\sim} \text{Normal}(\theta, (\sqrt{\theta})^2)$, with variance θ . Find the UMVUE of $\theta(1+\theta)$.

The joint pdf of $x_1, \dots, x_n$,

$$\begin{aligned} f_{\theta}(x_1, \dots, x_n) &= \exp \left\{ -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \theta - \frac{1}{2\theta} \sum_{i=1}^n (x_i - \theta)^2 \right\} \\ &= \exp \left\{ \underbrace{-\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \theta - \frac{n\theta}{2}}_{A(\theta)} + \underbrace{\sum_{i=1}^n x_i}_{B(\underline{x})} - \underbrace{\frac{1}{2\theta} \sum_{i=1}^n x_i^2}_{C(\theta)T(\underline{x})} \right\} \end{aligned}$$

Thus, $N(\theta, (\sqrt{\theta})^2)$ belongs to the exponential family and

$T(\underline{x}) = \sum_{i=1}^n x_i^2$ is a complete-sufficient statistic (css) of θ .

$$E(T(\underline{x})) = \sum_{i=1}^n E(x_i^2) = n \{ \text{Var}(x_1) + E^2(x_1) \} = n \{ \theta + \theta^2 \}$$

$$\therefore E\left(\frac{1}{n} \sum_{i=1}^n x_i^2\right) = \theta(1+\theta).$$

As $\frac{1}{n} \sum_{i=1}^n x_i^2$ is a function of css and is unbiased for $\theta(1+\theta)$,

it is the UMVUE of $\theta(1+\theta)$.

5. Let X_n be the outcome of the n -th toss of a coin with $P(\{H\}) = 0.75$, $n \in \mathbb{N}$. Can you find the approximate probability that after $N = 900$ tosses at least 450 heads are observed?

Let $X_i = \begin{cases} 1 & \text{if the } i\text{th toss results in } \{H\} \\ 0 & \text{otherwise,} \end{cases} \quad i=1, \dots, 900 = N$ Then $X_i \sim \text{Bernoulli}(0.75)$, $i=1, \dots, N$
 We assume that $X_1, \dots, X_N$ are independently distributed.
 To find $P\left(\sum_{i=1}^N X_i \geq 450\right)$.
 As $E(X_i) = 0.75$ and $\text{var}(X_i) = 0.75 \times 0.25 = 3/16$, by CLT we have

$$Z_N = \frac{\sqrt{N}(\bar{X}_N - 3/4)}{\sqrt{3/4}} \xrightarrow{d} N(0, 1).$$

 Thus,
$$P\left(\sum_{i=1}^N X_i \geq 450\right) = P\left(\frac{\sqrt{N}(\bar{X}_N - 3/4)}{\sqrt{3/4}} \geq 30\left(\frac{1}{2} - \frac{3}{4}\right)\right) = P\left(Z_N \geq -\frac{30}{\sqrt{3}}\right) \approx 1 - \Phi(-17.32) \approx 1. \text{ [ANS]}$$

6. Let $X_1, \dots, X_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma^2)$. Choose an appropriate pivot $T_n(\sigma^2)$ based on the MLE of σ^2 , and the point a such that $P(T_n(\sigma^2) < a) = 1 - \alpha$. Hence find a one-sided $1 - \alpha$ -confidence interval for σ^2 .

The MLE of σ^2 is $\hat{\sigma}_{ML}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x}_n)^2$.
 We know that, $\frac{n \hat{\sigma}_{ML}^2}{\sigma^2} \sim \chi_{n-1}^2$, which is free of σ^2 .
 Thus, $T_n(\sigma^2) = \frac{n \hat{\sigma}_{ML}^2}{\sigma^2}$ is the chosen pivot, and $a = \chi_{\alpha; n-1}^2$, i.e.,
 a is the upper- α point of χ_{n-1}^2 distribution.
 Hence
$$P\left(T_n(\sigma^2) < \chi_{\alpha; n-1}^2\right) = 1 - \alpha$$

$$\Rightarrow P\left(\frac{n \hat{\sigma}_{ML}^2}{\chi_{\alpha; n-1}^2} < \sigma^2\right) = 1 - \alpha.$$

 Thus, $\left(\frac{n \hat{\sigma}_{ML}^2}{\chi_{\alpha; n-1}^2}, \infty\right)$ is the one-sided $(1 - \alpha)$ -confidence interval of σ^2 .

7. Consider the testing problem $H_0 : X \sim \text{exponential}(\lambda)$ against $H_1 : X \sim \text{Gamma}(k, \lambda)$ for some $k > 1$. Derive a most powerful level- α test for testing H_0 vs H_1 based on one sample.

Let f_0 and f_1 be the pdf of X under H_0 and H_1 . Then

$$f_0(x) = \lambda e^{-\lambda x} \quad \text{and} \quad f_1(x) = \frac{\lambda^k x^{k-1} e^{-\lambda x}}{\Gamma(k)}, \quad x > 0.$$

By NP lemma the MP test is of the form

$$\phi^*(x) = \begin{cases} 1 & \text{if } \frac{f_1(x)}{f_0(x)} = \lambda(x) > k_0 \\ 0 & \text{if } \lambda(x) < k_0 \end{cases} \quad ; \quad \text{and satisfies} \quad E_{H_0}(\phi^*(x)) = \alpha.$$

$$\text{Here } \lambda(x) = \frac{x^{k-1} \lambda^k}{\Gamma(k)} > k_0 \Leftrightarrow x > \left(\frac{k_0 \Gamma(k)}{\lambda^k} \right)^{1/k} = c_0.$$

$$\text{Thus, } \phi^*(x) = \begin{cases} 1 & \text{if } x > c_0 \\ 0 & \text{if } x \leq c_0 \end{cases} \quad (\text{as } P_{H_0}(x = c_0) = 0) \quad \text{--- (*)}$$

$$\begin{aligned} \text{Also, } E_{H_0}(\phi^*(x)) = \alpha &\Rightarrow P_{H_0}(x > c_0) = \int_{c_0}^{\infty} \lambda e^{-\lambda x} dx = \left[-e^{-\lambda x} \right]_{c_0}^{\infty} = e^{-\lambda c_0} = \alpha \\ &\Rightarrow c_0 = -\lambda^{-1} \log \alpha. \end{aligned}$$

Thus, the MP test is as in (*) with $c_0 = -\lambda^{-1} \log \alpha$.

Rough work

Rough work

Rough work